

Lab assignment 2 - Basic imaging operations and applying linear filters with OpenCV

Complete



8.9.2025

Attempt 1



Review Feedback

27.8.2025

Attempt 1 Score:

Complete



Add Comment

Anonymous Grading: no

Unlimited Attempts Allowed

25.8.2025 to 17.11.2025

Details

Lab assignment 2 (individual)

In this individual lab assignment, every student must complete the following tasks.

The code created for the exercises has to be delivered to the repository from Lab assignment 1.

SUBMISSION: The submission shall be a link to your git repository along with the outputs from part II.

I. Assignment directory

1. Create a new directory named "assignment_2" inside your "IKT213_lastname" repository.

In this exercise, we will use the image "[lena.png \(https://uia.instructure.com/courses/18389/files/3005644?wrap=1\)](https://uia.instructure.com/courses/18389/files/3005644?wrap=1)". [↓ \(https://uia.instructure.com/courses/18389/files/3005644/download?download_frd=1\)](https://uia.instructure.com/courses/18389/files/3005644/download?download_frd=1)". Use "cv2.imread(...)" to load the image!

II. Create functions for padding, cropping, resizing, copying, grayscale, hue-shift, HSV, smoothing, rotation

1. **Padding:** Create a border around the original image which reflects the edges of the original image.

To learn more about padding, go to "http://www.bim-times.com/opencv/3.3.0/d3/df2/tutorial_py_basic_ops.html" [↗](http://www.bim-times.com/opencv/3.3.0/d3/df2/tutorial_py_basic_ops.html) ("http://www.bim-times.com/opencv/3.3.0/d3/df2/tutorial_py_basic_ops.html").

- A. Function name: "padding"
- B. Function arguments: **image, border_width**
- C. Make a border with the reflection of the image with **border_width=100**
- D. Save the image

2. **Cropping:** Cut the image such that it returns only the part of the image which is of interest (in your case, the face on lena.png)

To learn more about cropping, go to "<https://learnopencv.com/cropping-an-image-using-opencv/>"  ("<https://learnopencv.com/cropping-an-image-using-opencv/>").

- A. Function name: **"crop"**
- B. Function arguments: **image, x_0, x_1, y_0, y_1**
- C. Crop the image by **80 pixels** from left and from the top, and **130 pixels** from the bottom and from the right
- D. Save the image

3. **Resize:** Create a function which lets you resize the image

To learn more about resizing, go to "<https://www.geeksforgeeks.org/image-resizing-using-opencv-python/>"  ("<https://www.geeksforgeeks.org/image-resizing-using-opencv-python/>").

- A. Function name: **"resize"**
- B. Function arguments: **image, width, height**
- C. Resize the image to **200x200 pixels**
- D. Save the image

4. **Manual copy:** use what we learned in Lab 1 to get the width, height, and channels, and create a new empty array with `"emptyPictureArray = np.zeros((height, width, 3), dtype=np.uint8)"`, and fill it with the pixels from the Lena image.

Do not use the cv2 copy() function

- A. Function name: **"copy"**
- B. Function arguments: **image, emptyPictureArray**
- C. Copy the pixels from the image to the empty image
- D. Save the image

5. **Grayscale:** Convert the colored image to a grayscale image.

to learn more about grayscale, go to "<https://techtutorialsx.com/2018/06/02/python-opencv-converting-an-image-to-gray-scale/>"  ("<https://techtutorialsx.com/2018/06/02/python-opencv-converting-an-image-to-gray-scale/>").

- A. Function name: **"grayscale"**
- B. Function argument: **image**
- C. Make the image grayscale
- D. Save the image

6. **HSV:** Convert an RGB image to use HSV

to learn more about HSV, go to "<https://techtutorialsx.com/2019/11/08/python-opencv-converting-image-to-hsv/>"  ("<https://techtutorialsx.com/2019/11/08/python-opencv-converting-image-to-hsv/>").

[converting-image-to-hsv/](#)".

- A. Function name: **"hsv"**
- B. Function argument: **image**
- C. Convert the image to hsv
- D. Save the image

7. Color shifting: Shift the color values of a given RGB image

we have learned that an image is essentially a three-dimensional array. Now use this knowledge to change the color value of an image.

- A. Function name: **"hue_shifted"**
- B. Function arguments: **image, emptyPictureArray, hue**
- C. Shift the image **color value by 50** for all color values
(Remember that the color values go from 0 to 255, what will happen to the color when the value goes beyond 255 or below 0?)
- D. Save the image

8. Smoothing: You will be smoothing the original image; adding a blur to the image

to learn more about filters and smoothing, go to "<https://analyticsindiamag.com/complete-tutorial-on-linear-and-non-linear-filters-using-opencv/>" [↗](https://analyticsindiamag.com/complete-tutorial-on-linear-and-non-linear-filters-using-opencv/%C2%A0) (<https://analyticsindiamag.com/complete-tutorial-on-linear-and-non-linear-filters-using-opencv/%C2%A0>) or <https://www.tutorialkart.com/opencv/python/opencv-python-gaussian-image-smoothing/> [↗](https://www.tutorialkart.com/opencv/python/opencv-python-gaussian-image-smoothing/) (<https://www.tutorialkart.com/opencv/python/opencv-python-gaussian-image-smoothing/>)".

- A. Function name: **"smoothing"**
- B. Function argument: **image**
- C. Blur the image by using the kernel size **kernel_size=(15,15)**, and "default border"
- D. Save the image

9. Rotation: You will be rotating the image

To learn more about rotation, go to "<https://www.geeksforgeeks.org/python-opencv-cv2-rotate-method/>" [↗](https://www.geeksforgeeks.org/python-opencv-cv2-rotate-method/) (<https://www.geeksforgeeks.org/python-opencv-cv2-rotate-method/>)".

- A. Function name: **"rotation"**
- B. Function arguments: **image, rotation_angle**
- C. If the input variable **rotation_angle** is **90**, rotate the image **90 degrees clockwise**
- D. If the input variable **rotation_angle** is **180**, rotate the image **180 degrees**
- E. Save the image

Do you need help in the lab or in Discord? Find links here: [Lab Information \(https://uia.instructure.com/courses/18389/pages/lab-information\)](https://uia.instructure.com/courses/18389/pages/lab-information)
